

Geographical Information Systems نظم المعلومات الجغرافية

Prepared by

Dr.

Nermeen L. Labeeb

© Prentice
Hall, 2002

Fundamentals of GIS: Map Elements & Design

The Principles of Spatial Visualization
and Communication



1 PART

Fundamentals of GIS

CHAPTER

2

Map: Elements & Design

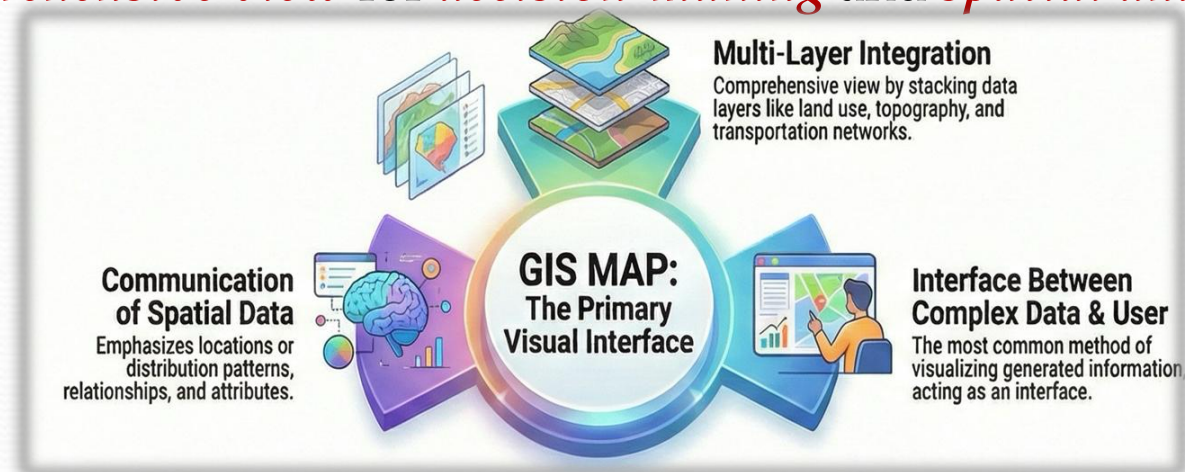
By the end of this chapter you should be able to:

- 🌐 Explain what *Map* is.
- 🌐 Describe the basic *elements* of a map.
- 🌐 Mention the different *expressions* of *map scale*.
- 🌐 Illustrate what is meant by *features generalization*.
- 🌐 Discuss the relationship between the *map scale* and *features generalization*.
- 🌐 Explore the different types of maps.

2.1

WHAT A MAP IS?

- A **Map**, As an **interface** to a (**GIS**), and the most common method of **visualizing** information generated by it, is the most vital part we have to consider it in our study of GIS.
- In (GIS), a **map** serves as the primary **visual representation** and **communication** of **spatial data**; whether the emphasis is on the **location** or the **distribution pattern** of the data.
- It displays **geographical features, relationships, and attributes**, allowing users to **analyze** and **interpret** information about a **specific area**.
- Maps in GIS integrate various **layers of data**, such as **roads, land use, and topography**, providing a **comprehensive view** for **decision-making** and **spatial analysis**.



The Map as a Data Interface

In GIS, a **map** is the primary visual representation of spatial data, acting as an interface to visualize distribution patterns and locations.

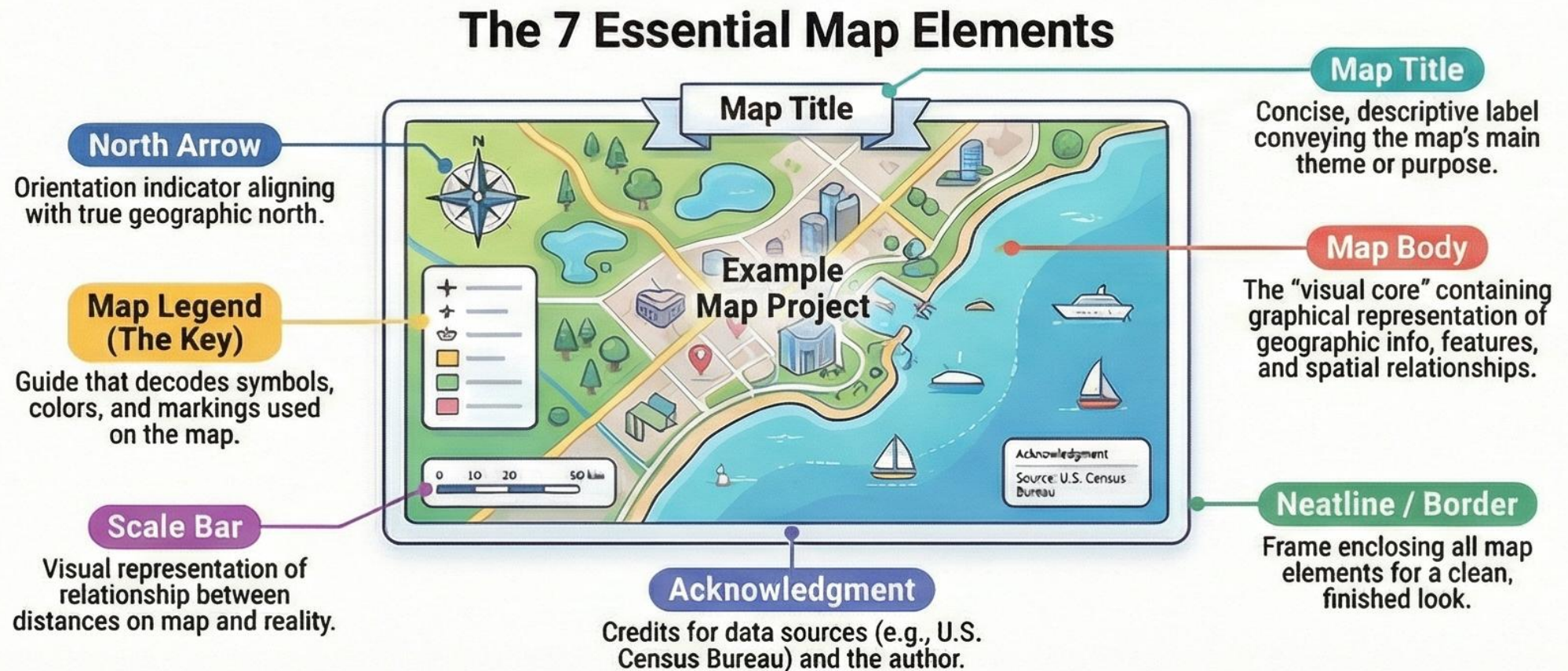
- **Integration:** Combines layers of data (Roads, Land Use, Topography).
- **Analysis:** Enables interpretation of relationships for decision-making.
- **Communication:** Translates complex features into a comprehensive view.



2.2

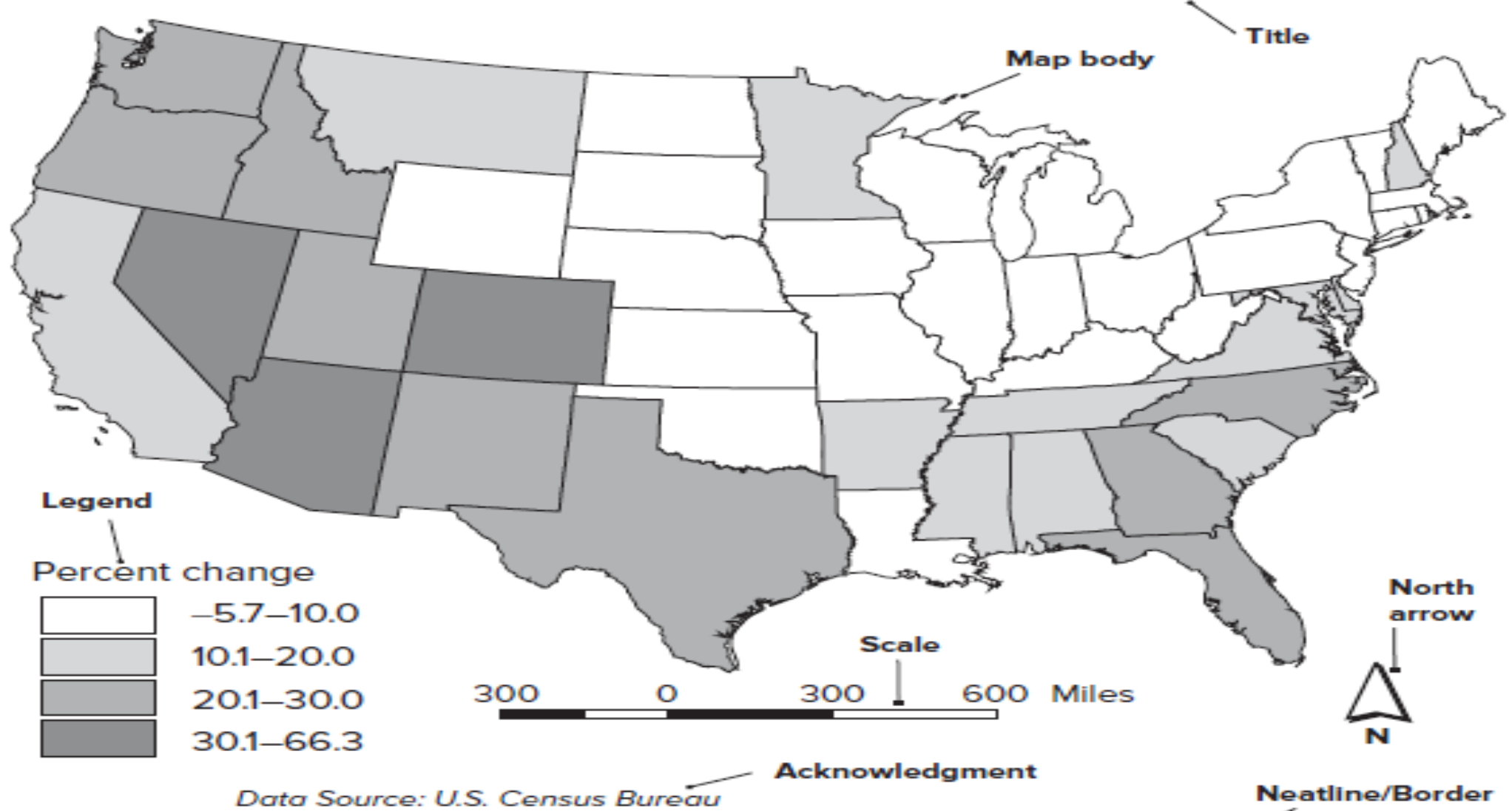
MAP ELEMENTS

- Common map elements are the *title*, *body*, *legend*, *north arrow*, *scale bar*, *acknowledgment*, and *neatline / map border* (Figure 1_a, b, c).



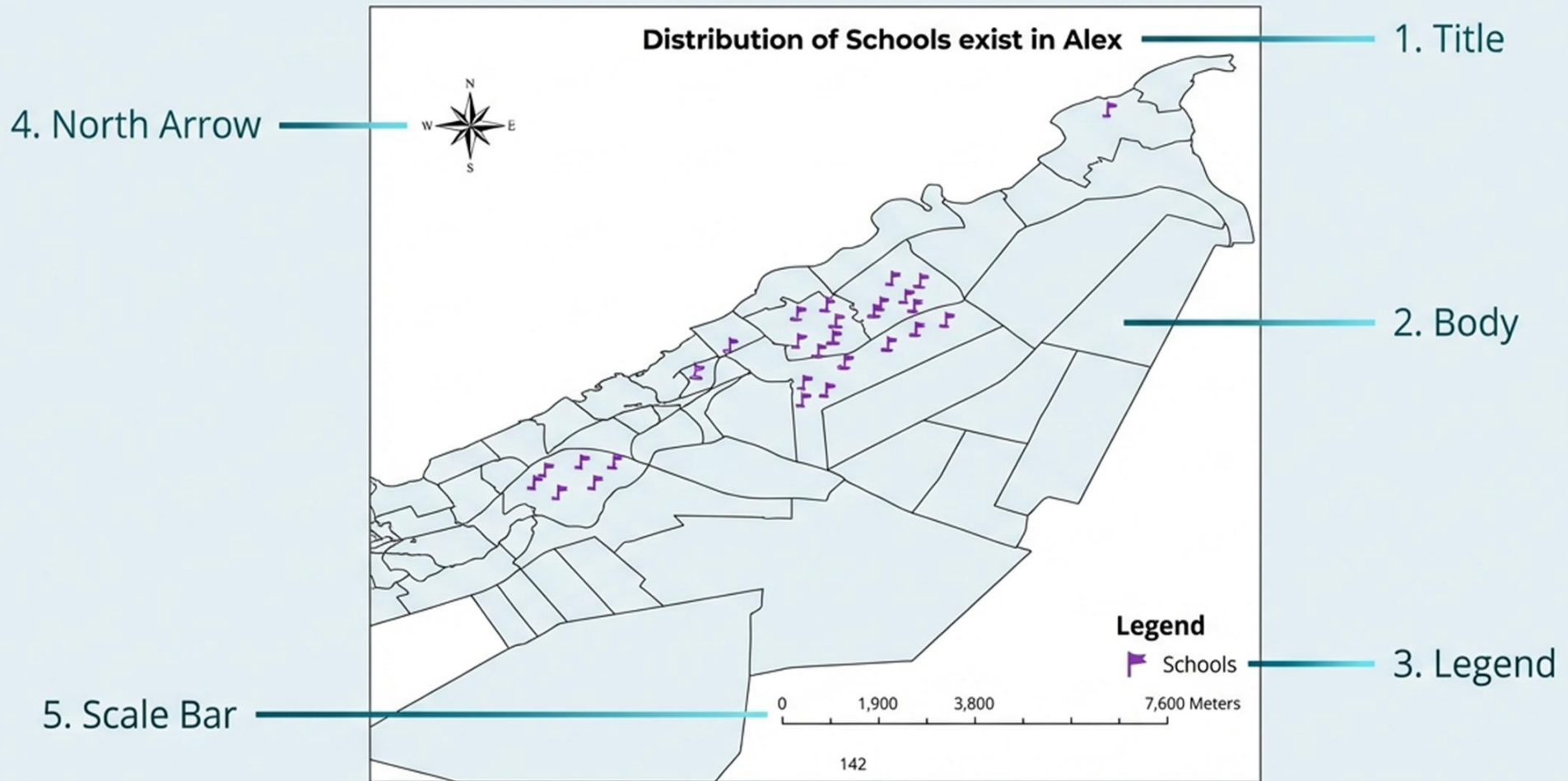
(Figure 1-a) Common map elements

Population Change by State, 1990–2000



(Figure 1-b) Common map elements

The Anatomy of an Effective Map

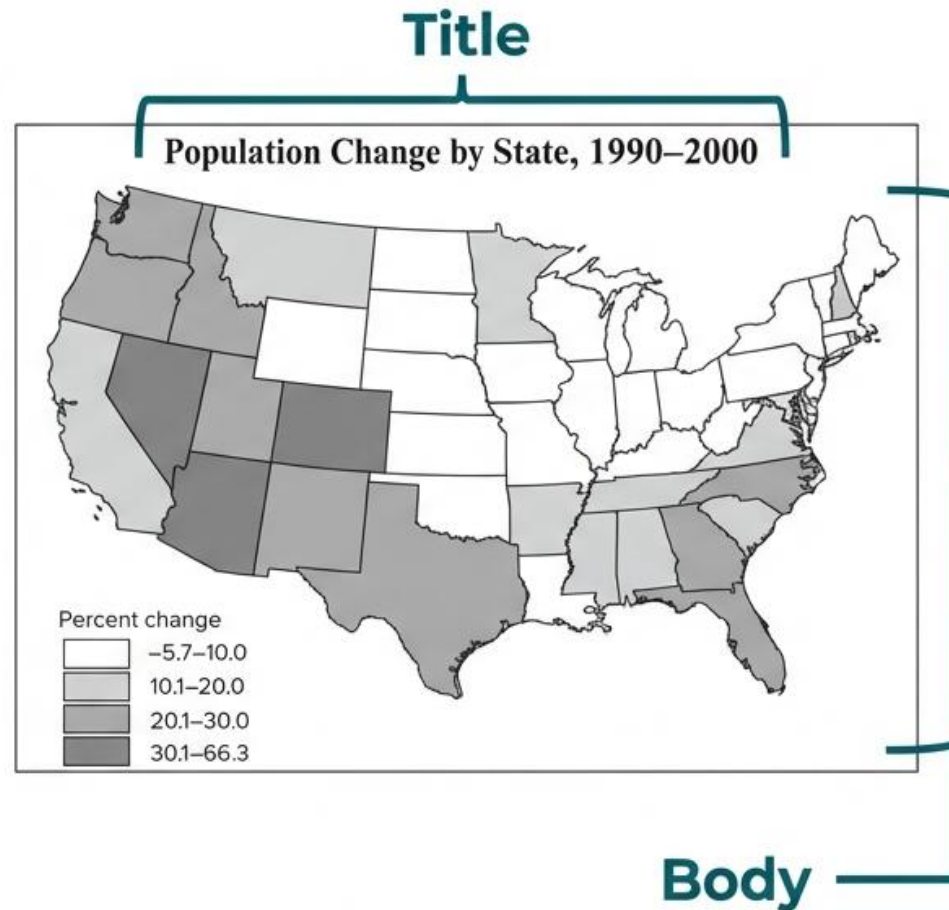


(Figure 1-c) Common map elements

Establishing Context and Core

Component 1: The Map Title

- **Function:** A concise, descriptive headline.
- **Purpose:** Captures attention and provides the essential “what” and “where”.



Component 2: The Map Body

- **Function:** The central, substantive portion containing the graphical representation.
- **Purpose:** The visual core holding spatial data layers required for analysis.

(Figure 2) Map context and core

1. Map Title:

- a *concise* and *descriptive* text element **prominently** **بشكل واضح** displayed on a map to convey the main *theme, purpose*, or *subject matter* of the *map*.
- It serves as a *brief* and *informative label* that helps users understand the *content* and *context* **سياق** of the map, providing essential information about the *geographical area* or *topic* being represented.
- The map title is typically positioned at the *top* or *center* of the map, and it is designed to capture the viewer's attention and guide them in interpreting the information presented on the map.

2. Map Body:

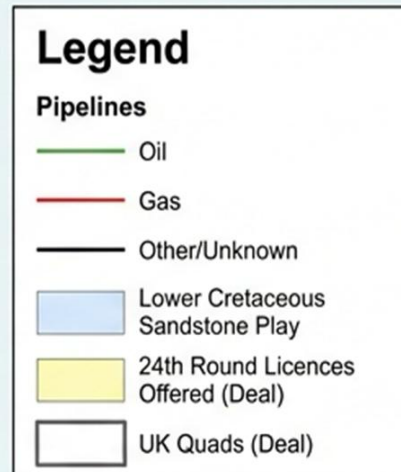
- refers to the *central* and *substantive* **الحقيقي** portion of a map that contains the *graphical representation* of *geographic information*.
- It includes various *layers* of *spatial data*, such as **roads, rivers, boundaries**, and other *features*, organized to visually convey information about the *physical* or *thematic characteristics* of a specific area.
- The *map body* constitutes the *visual core* of the *map* and is crucial for users to interpret and analyze *spatial relationships, patterns*, and *attributes*.



Decoding Symbols and Orientation

Component 3: The Legend

Also known as the Key. Explains symbols, colors, and markings essential for decoding data.



Component 4: The North Arrow

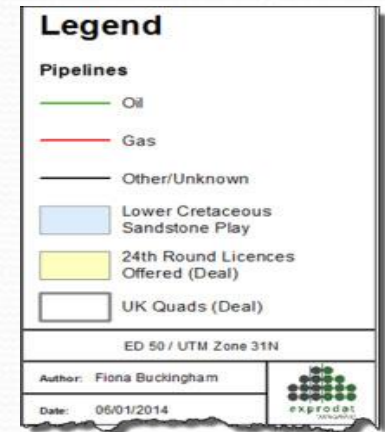
Indicates orientation relative to True North. Critical for accurate interpretation of spatial relationships.



(Figure 3) Map decoding and Orientation

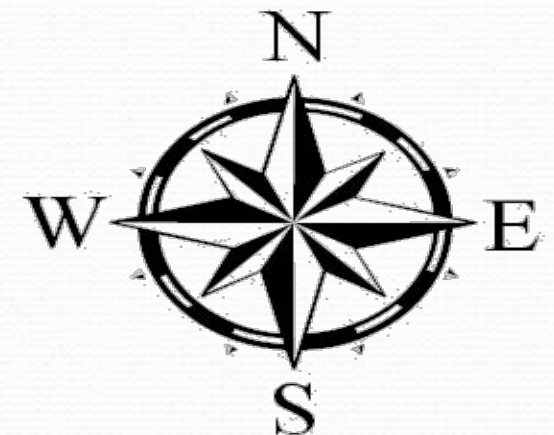
3. Map Legend:

- also known as a *key*, is a *graphical* or *written* guide accompanying a map that explains the *symbols*, *colors*, and other *graphical elements* used on the map.
- It provides a comprehensive explanation of the meaning and significance of the various *symbols* and *markings* employed to represent different *features*, *themes*, or *data categories* on the map.
- The *legend* aids map readers in *interpreting* the map by clarifying the representation of *geographical information* and ensuring a clear understanding of the *symbols* and their corresponding *attributes*.
- Typically positioned in a **designated area** on the map, the *legend* serves as an essential *reference tool* for users to *decode* the *visual information* presented on the map.



4. North Arrow:

- also known simply as *north indicator*, is a *symbol* or *graphic element* on a *map* that indicates the *orientation* of the map in relation to the *true north direction*.
- It provides users with a *visual reference point*, allowing them to align the *map* correctly with the actual *geographic cardinal directions*.
- It is particularly important because *maps* can be *oriented* in various *directions*, and the *north arrow* ensures that users can accurately interpret the *spatial relationships* and *orientations of features* on the map.
- It typically points towards the *geographic north*, but it may also include **additional directional information**, such as *cardinal points* (north, south, east, west) or *intermediate directions*.

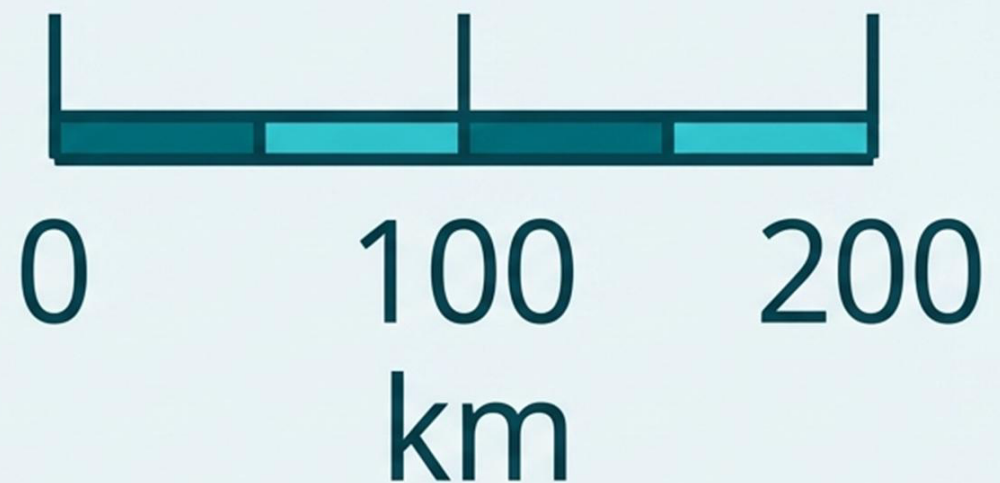


The Metric of Reality

Understanding Map Scale

The Scale Bar visually represents the relationship between map distance and real-world distance.

- **Linear Scale:** Graphic element divided into units (km/miles).
- **Estimation:** Allows users to gauge physical measurements.
- **Placement:** Typically positioned along the map's edge.



(Figure 4) Understanding map scale

Three Expressions of Scale

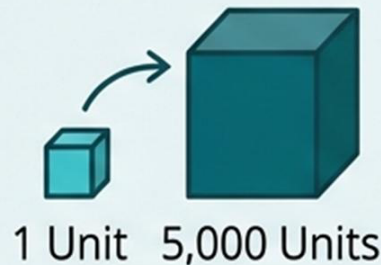
Ratio Scale

Mathematical ratio.



Example:

- 1:5,000 (1 unit on map = 5,000 units in reality).



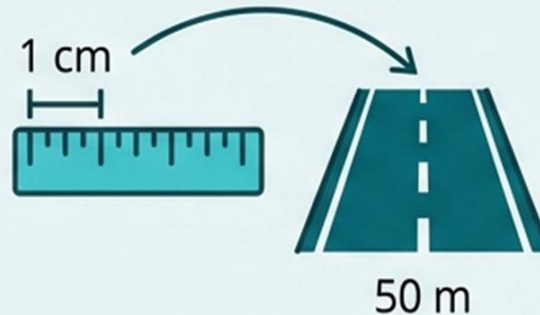
Verbal Scale

Descriptive language.



Example:

- 1 cm represents 50 m.



Graphical Scale

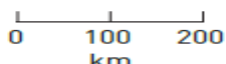
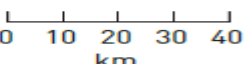
Visual line for estimation.



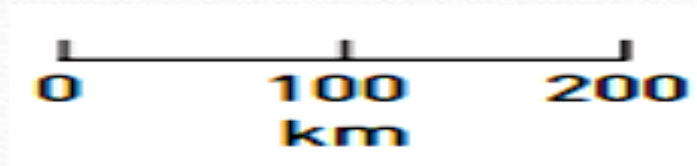
(Figure 5) Different expressions of map scale

5. Scale Bar:

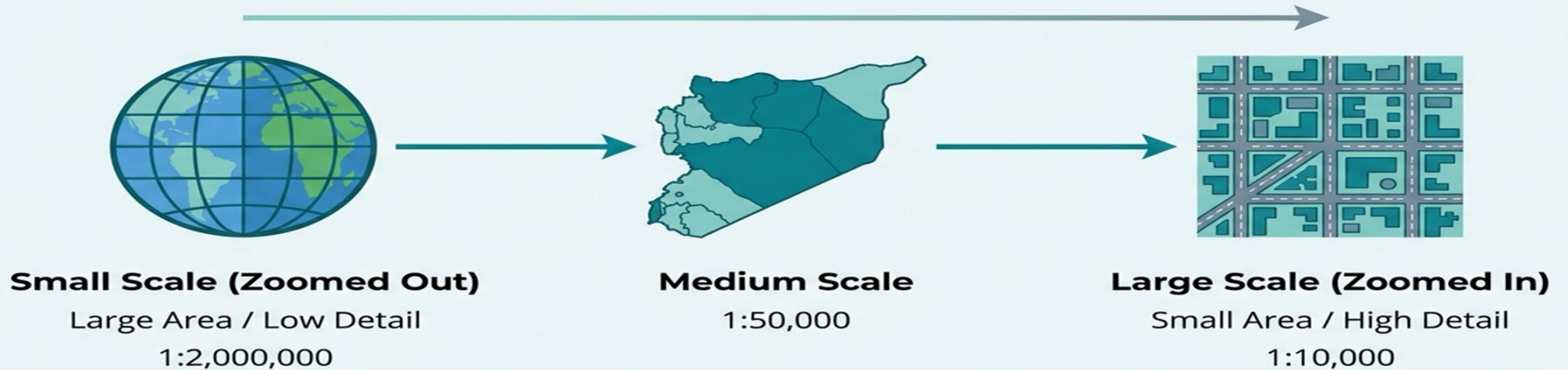
- It is a *graphic* element on a *map* that visually represents the *relationship* between the *distances on the map* and the *corresponding distances in the real world*.
- It is a *linear scale* that provides a means for users to gauge the actual physical measurements of *features* depicted on the *map*.
- It is typically placed along one of the map's edges or corners and is divided into *units of measurement*, such as *kilometers*, *miles*, or both.
- By using the map *scale bar*, map readers can estimate the *size* and *distance* between *geographical features* accurately.
- The *length* of the *scale bar* corresponds to a *specific measurement* in the *real world*, allowing users to make *informed decisions* about *distances* and *proportions* on the map.
- It is an essential tool for *spatial interpretation* and is crucial for understanding the *spatial relationships* presented on the map.

Ratio	1:5000	1:1,000,000
Verbal (nominal)	1 cm represents 50 m	1 cm represents 10 km
Graphical		

- Different *Expressions* of *Scale* can be used ; *Ratio*, *Verbal*, or *Graphical*. (Figure 2)
 - **Ratio scale**: conveys the scale information using the form of a **ratio**.
 - **Ex1**: 1 unit on the map represents 5,000 units in the real world.
 - **Ex2**: 1 unit on the map represents 1,000,000 units in the real world.
 - **Verbal scale**: conveys the scale information using **descriptive language**.
 - **Ex1**: "1 Cm on the map represents 50 m in reality."
 - **Ex2**: "1 Cm on the map represents 50 Km in reality."
 - **Graphical scale**: conveys the scale information using a *line* or *bar* divided into *segments*, with each segment labeled to indicate the *measurement unit* (e.g., kilometers, miles).
 - The *line bar* represents the *distance on the map*, and the *numbers* or *labels* indicate *the corresponding real-world distances*.



The Scale-Detail Spectrum

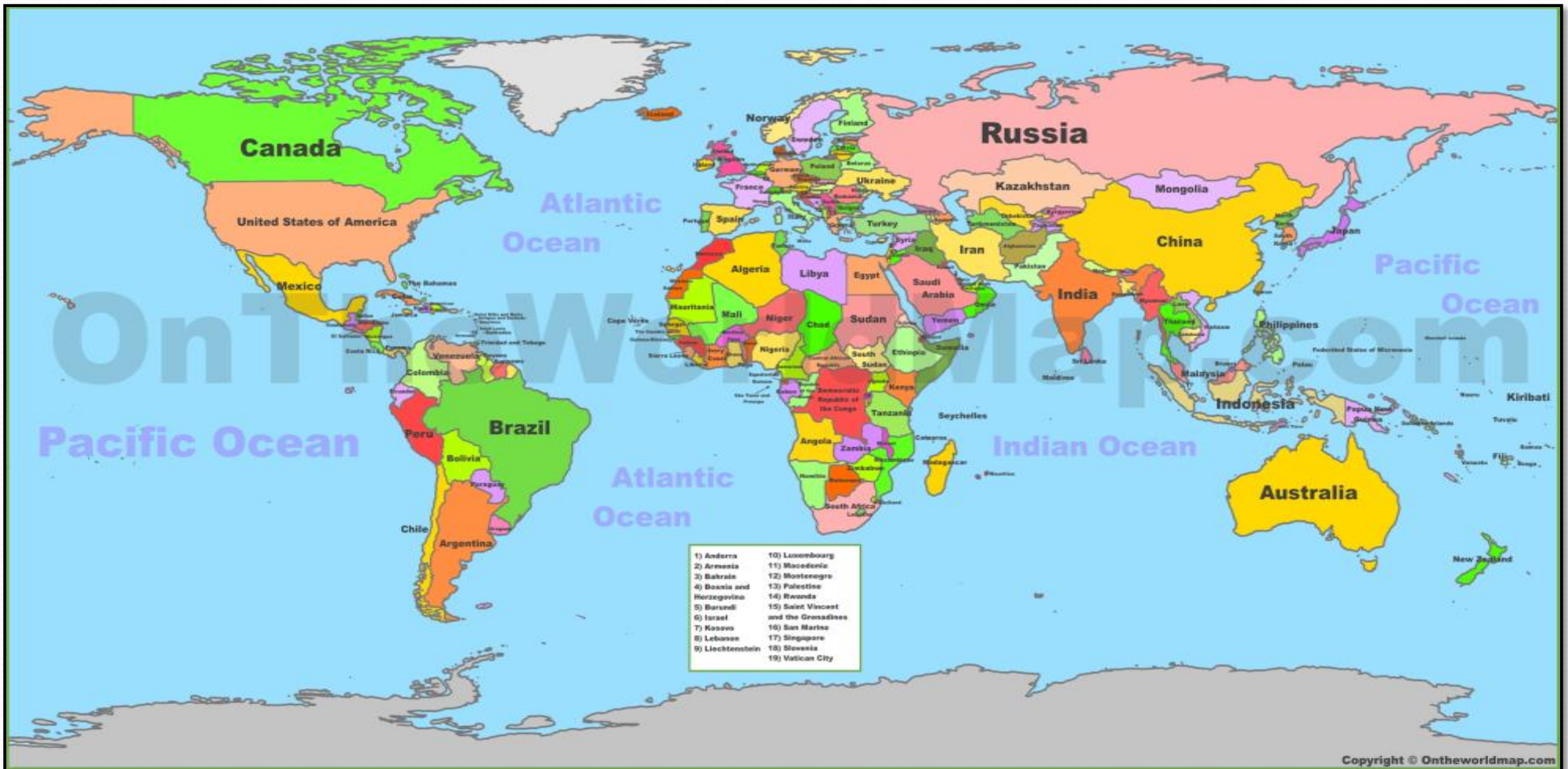


(Figure 6) Different Sizes of map scale

- **Map scales** can vary in **size** depending on the **level of details** needed for a particular map and **the extent of the geographic area** being represented.
- Here are different **sizes** or **types** of **map scales** based on their **level of detail**:
 1. **Small Scale**
 2. **Medium Scale**
 3. **Large Scale.**

1. **Small** Scale:

- Represents a **large geographic area** with **less details**. (Zoom out)
- Example: **World maps, continental maps.**
- Representative ratio is a small ratio (e.g., 1:2,000,000).



1. Medium Scale:

- Offers a **moderate level of detail** for a specific **region**.
- Example: **Country** or **regional** maps
- Representative ratio is a moderate ratio (e.g., 1:50,000).



1. Large Scale:

- Represents a *smaller geographic area* but with *high details*. (Zoom in)
- Example: **City** maps, **neighborhood** maps. خرائط الأحياء
- Representative ratio is a large ratio (e.g., 1:10,000)



The relationship between the **map scale** and **features Generalization**

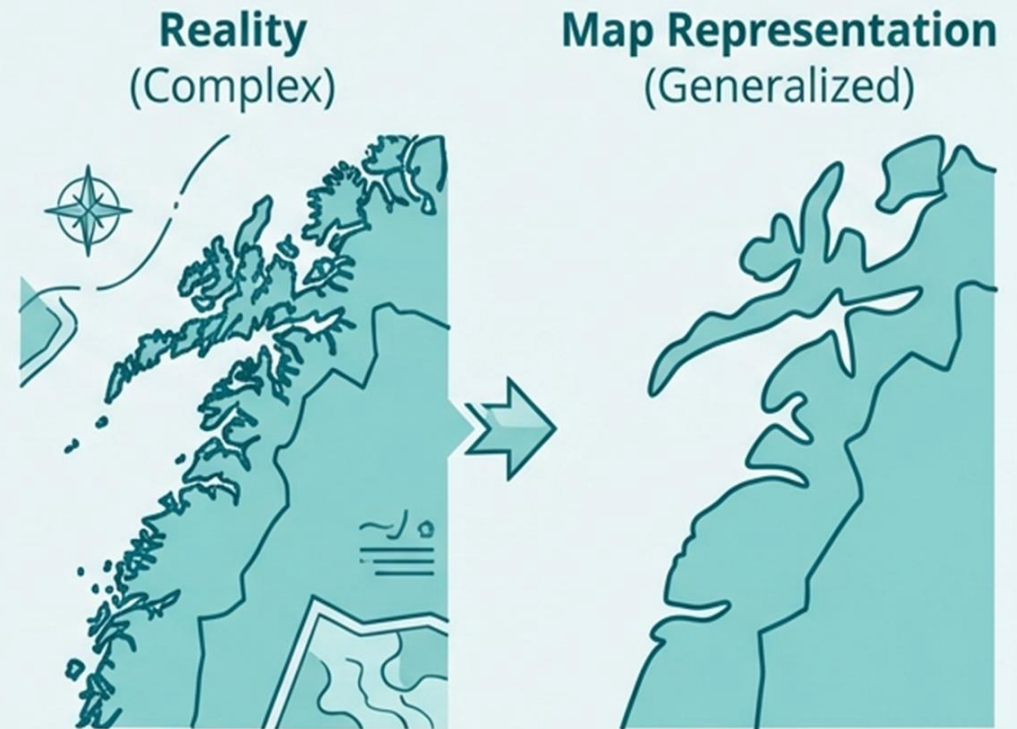
- *Feature generalization* refers to the process of *simplifying* and *abstracting geographic features* on a *map* to effectively represent them at *different scales*.
- As maps depict the *Earth's surface*, which is **complex** and **detailed**, it is often necessary to *generalize features* to improve *readability*, *reduce clutter*, and *convey essential information*.
- *Generalization* involves *making decisions* about what *information* to include, emphasize, or simplify based on the *map's purpose* and the *scale* at which it will be viewed.
- The relationship between *map scale* and *feature generalization* is a fundamental aspect of *cartography*.
- As the scale *decreases*, features are often *generalized* to maintain *clarity* and *effectively communicate the intended information*.
- The following examples illustrate how *feature generalization* varies based on the *map's scale* and *purpose*.

The Art of Feature Generalization

Definition: The process of simplifying and abstracting geographic features to effectively represent them at different scales.

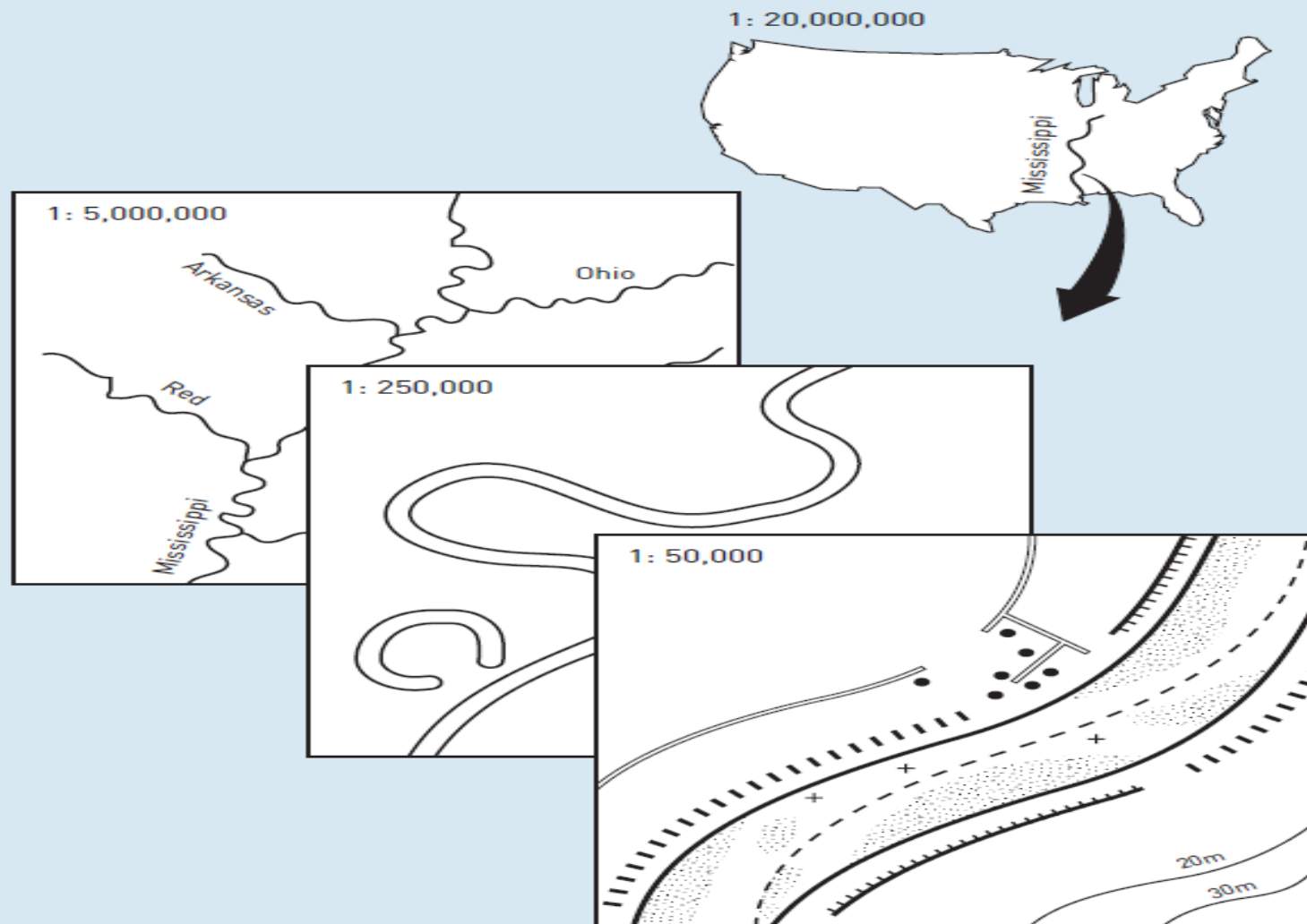
Why Generalize?

- Improve readability.
- Reduce visual clutter.
- Convey essential information.



(Figure 7) Feature Generalization

Example_01

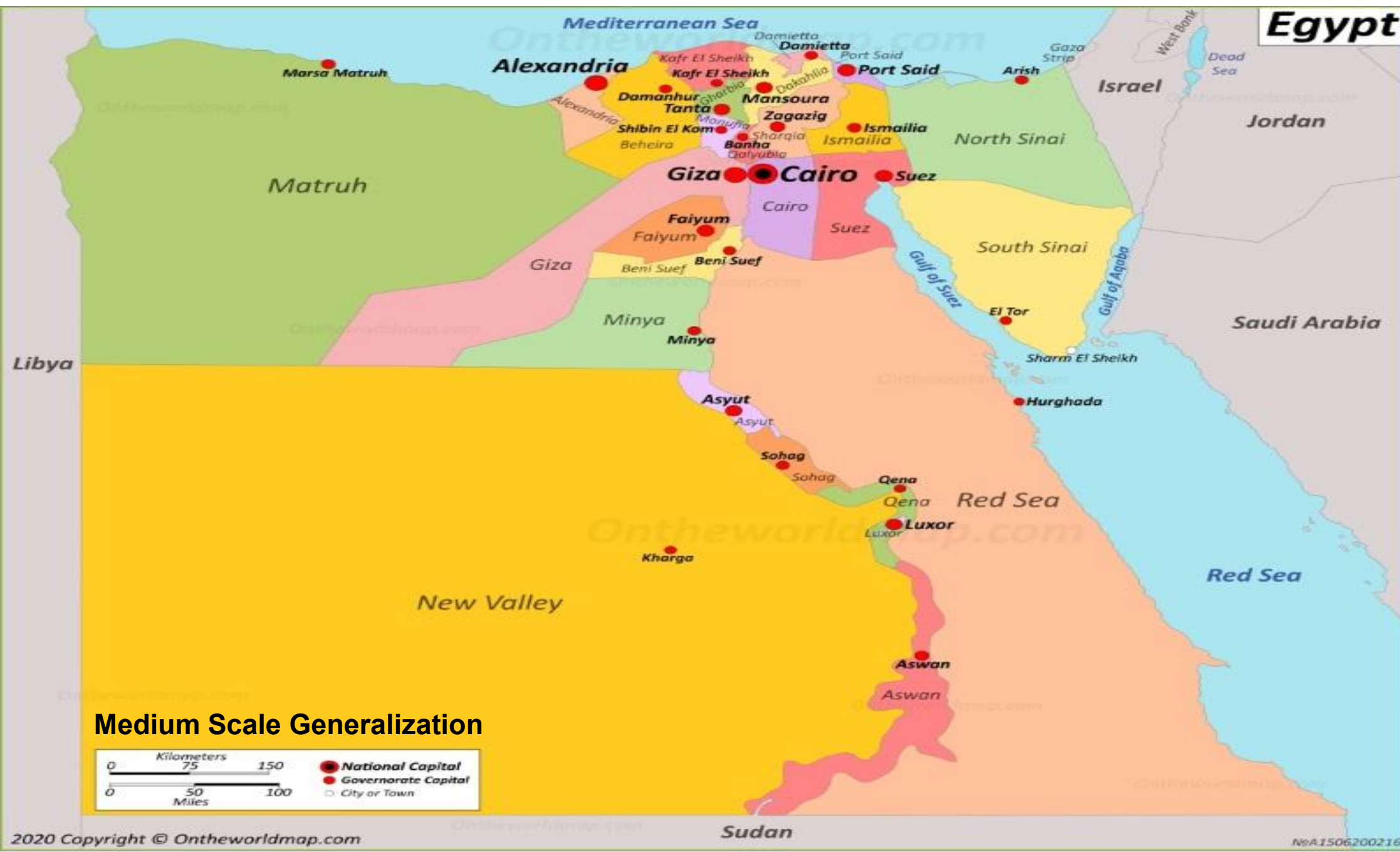


(Figure 8) Scale Related Generalization

Example_02

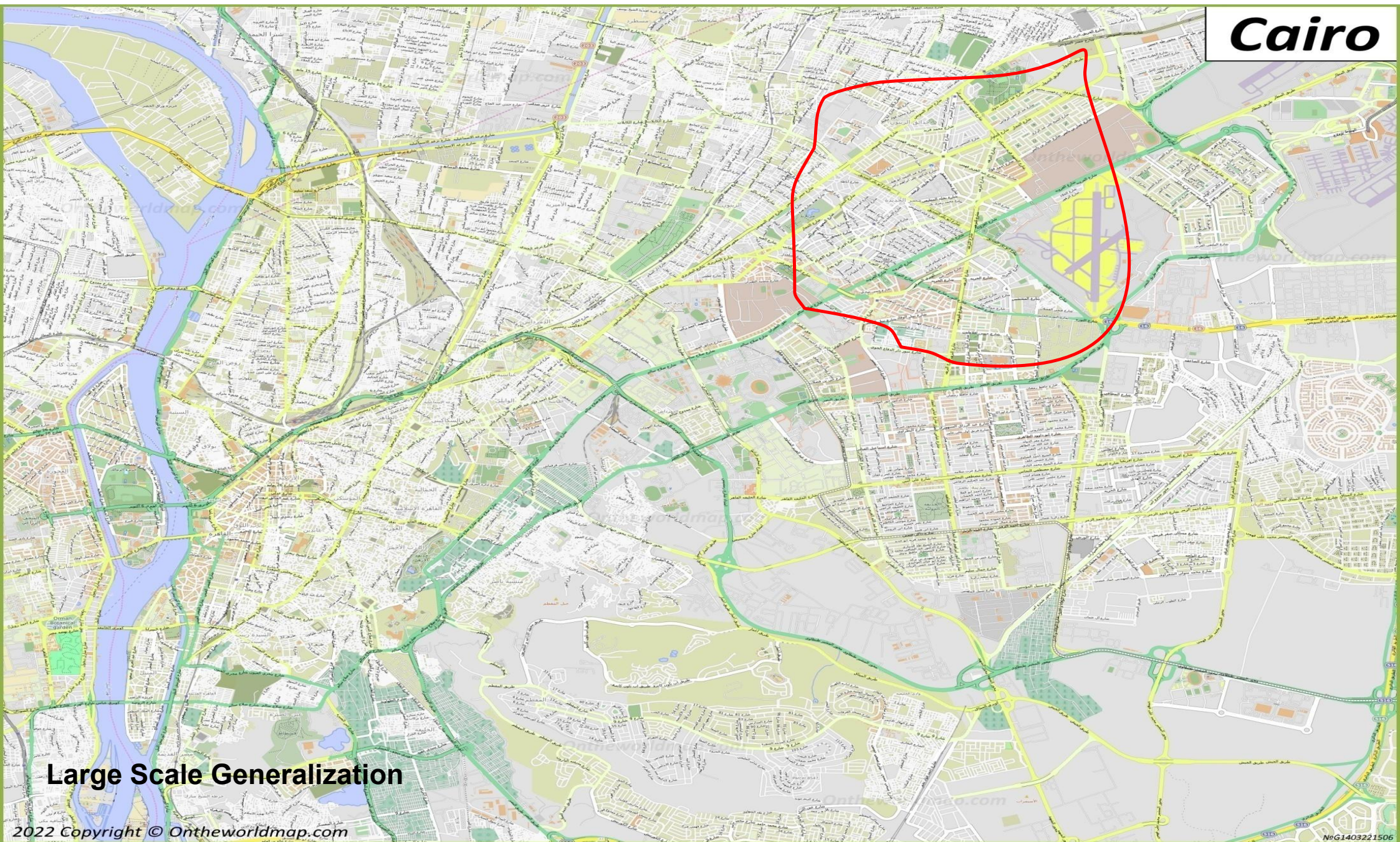


Egypt



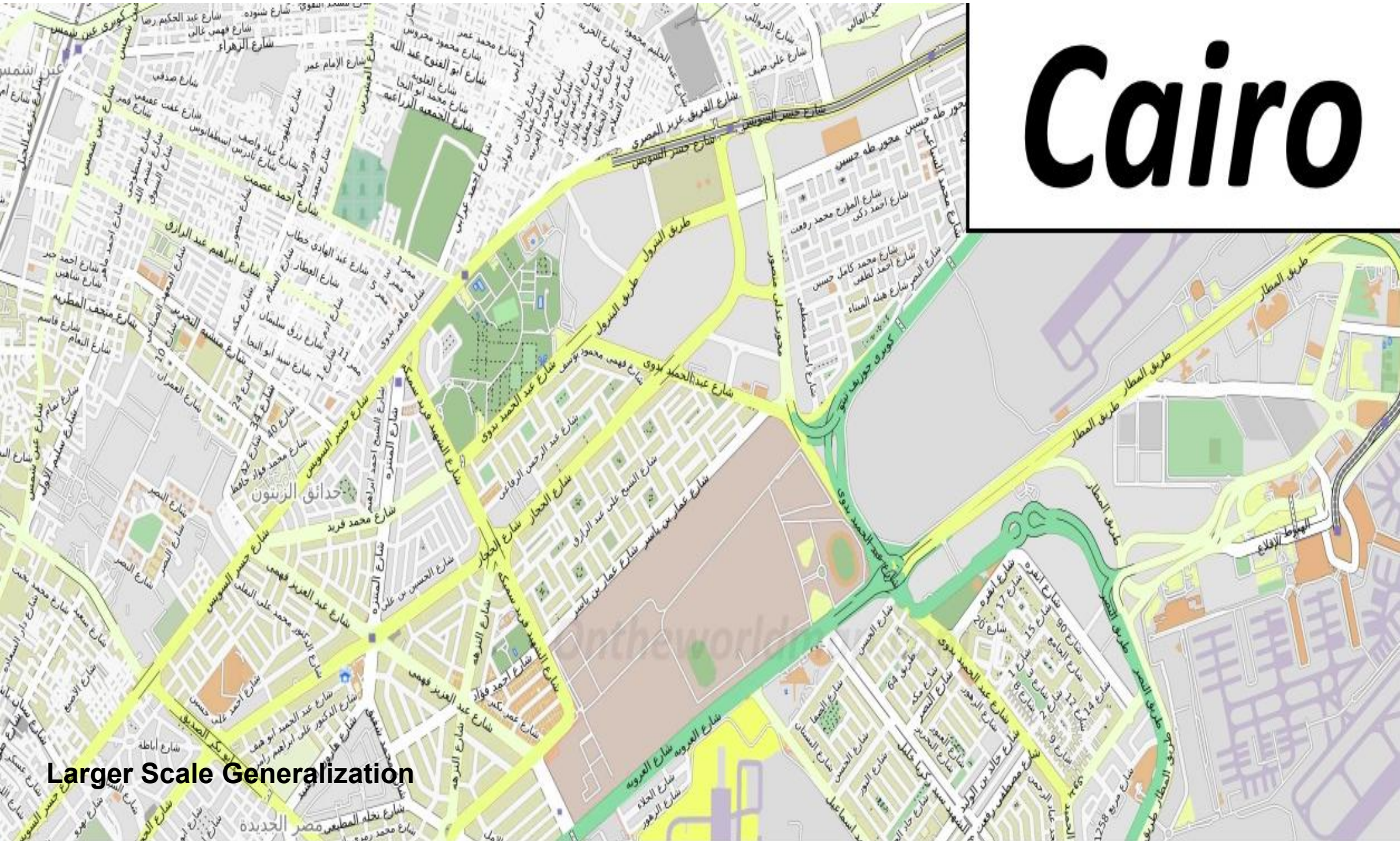
Medium Scale Generalization

Cairo

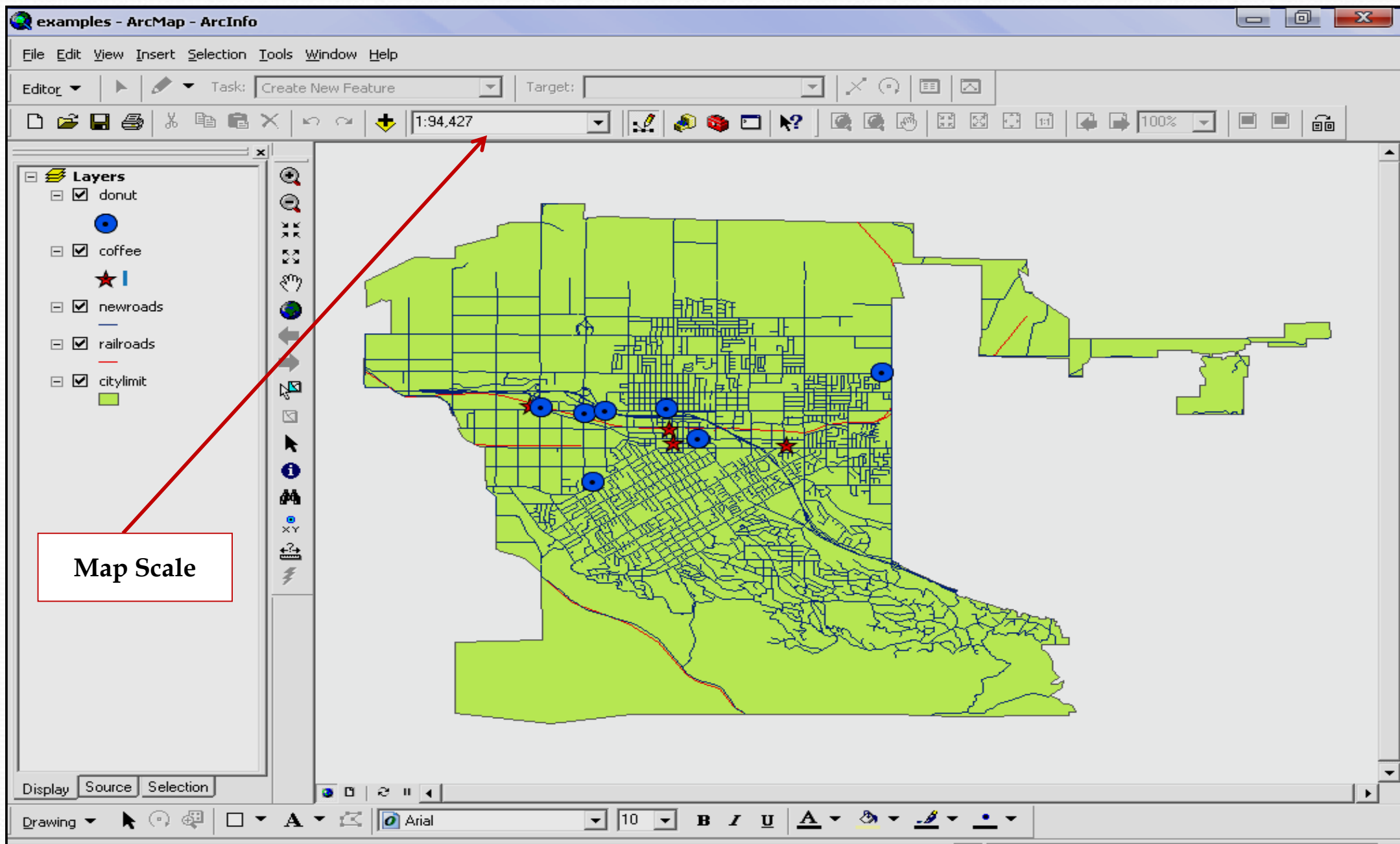


Large Scale Generalization

Cairo



Larger Scale Generalization



2.3

MAP TYPES

- (GIS) can represent various types of *maps*, each serving specific *purposes* and displaying *different types of information*.
- Generally, There are two types of maps represented in GIS:

1. General reference maps:

- are *comprehensive maps* that provide a **broad overview** of *geographic features* and *locations*.
- They aim to present a **variety of information**, often including political, physical, cultural, and economic aspects of a *region* or the *entire world*.
- These maps serve as foundational *references* for a wide range of *purposes* and are not focused on a specific *theme* or *topic*.

2. Thematic maps:

- are specialized maps designed to convey specific information about a particular theme, topic, or subject.
- Unlike general reference maps, thematic maps focus on representing data related to a specific subject or concept.

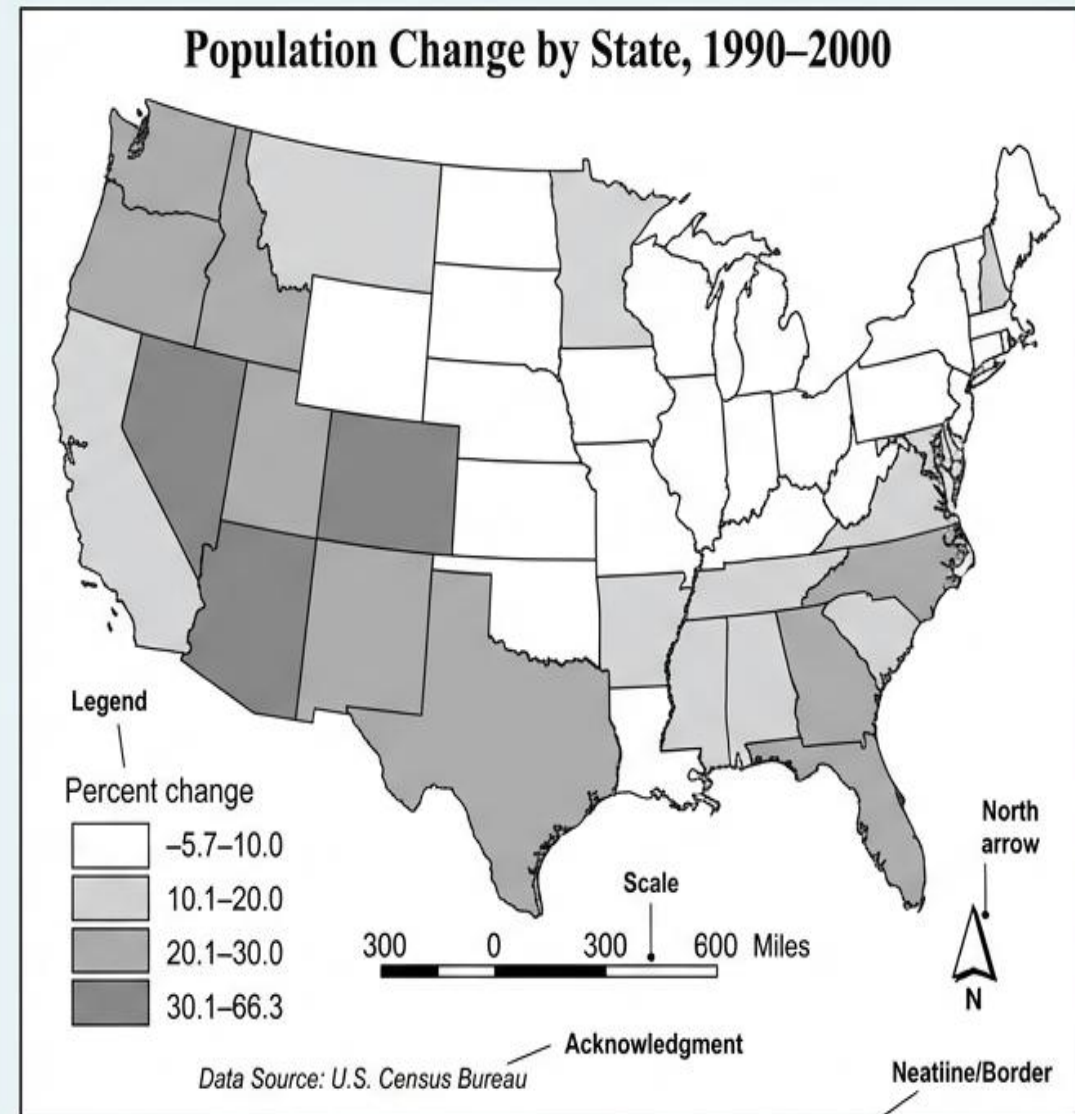
Category I: General Reference Maps

- **Purpose:** Comprehensive maps providing a broad overview of features and locations.
- **Characteristics:**
 - Foundational references not focused on a single theme.
 - Includes political, physical, cultural, and economic aspects.
- **Sub-Type:** Physical Maps focus on natural geography (mountains, rivers) using color to differentiate elevation.



Category II: Thematic Maps

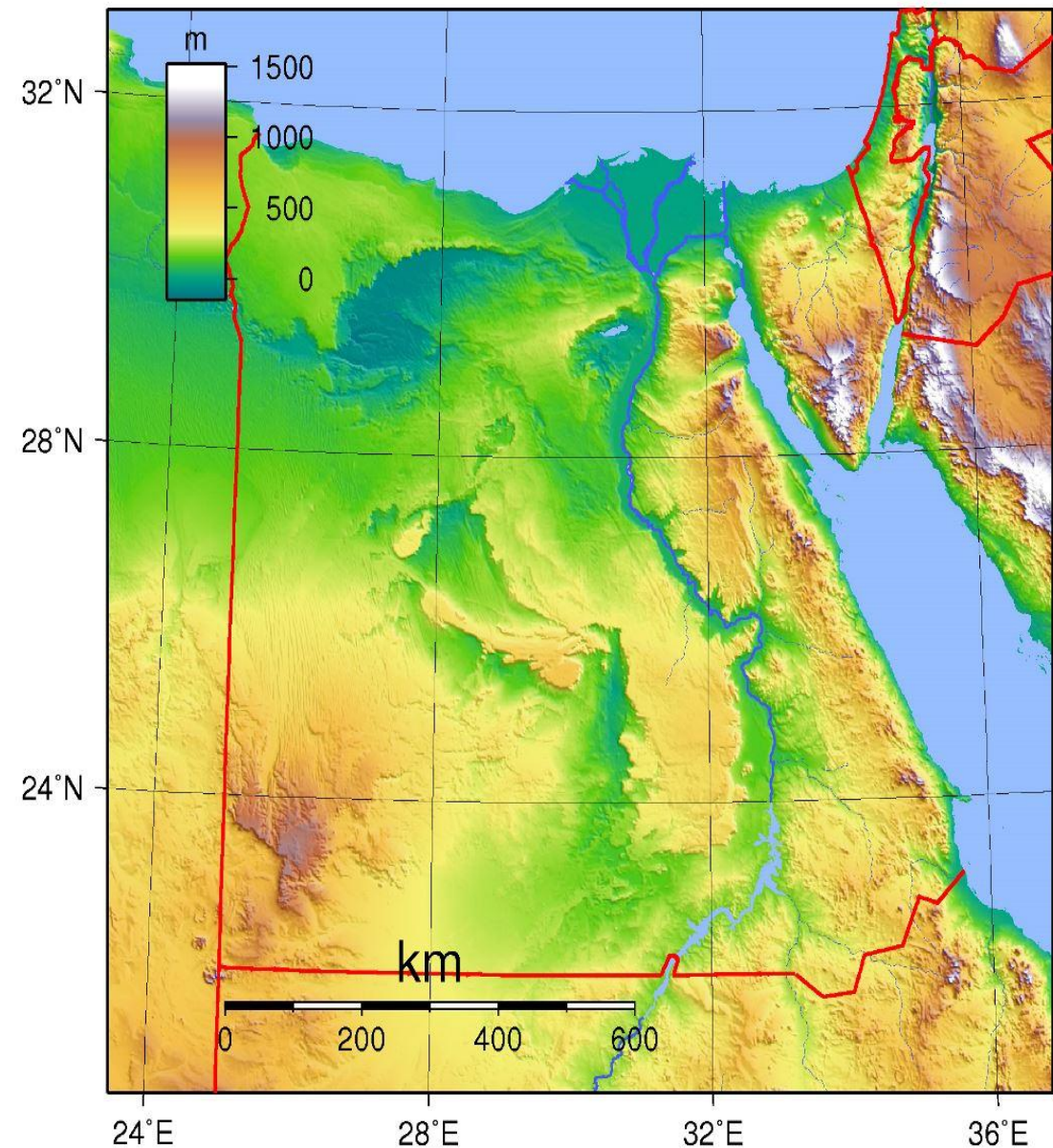
- **Purpose:** Specialized maps designed to convey specific information about a particular topic.
- **Characteristics:**
 - Focuses on data distribution related to a single concept.
 - Answers “What is the pattern?” rather than just “Where is it?”
- **Sub-Type:** Economic Maps display resource arrangement and activity.



- Types of **General reference maps** :

- 1. Physical Maps:

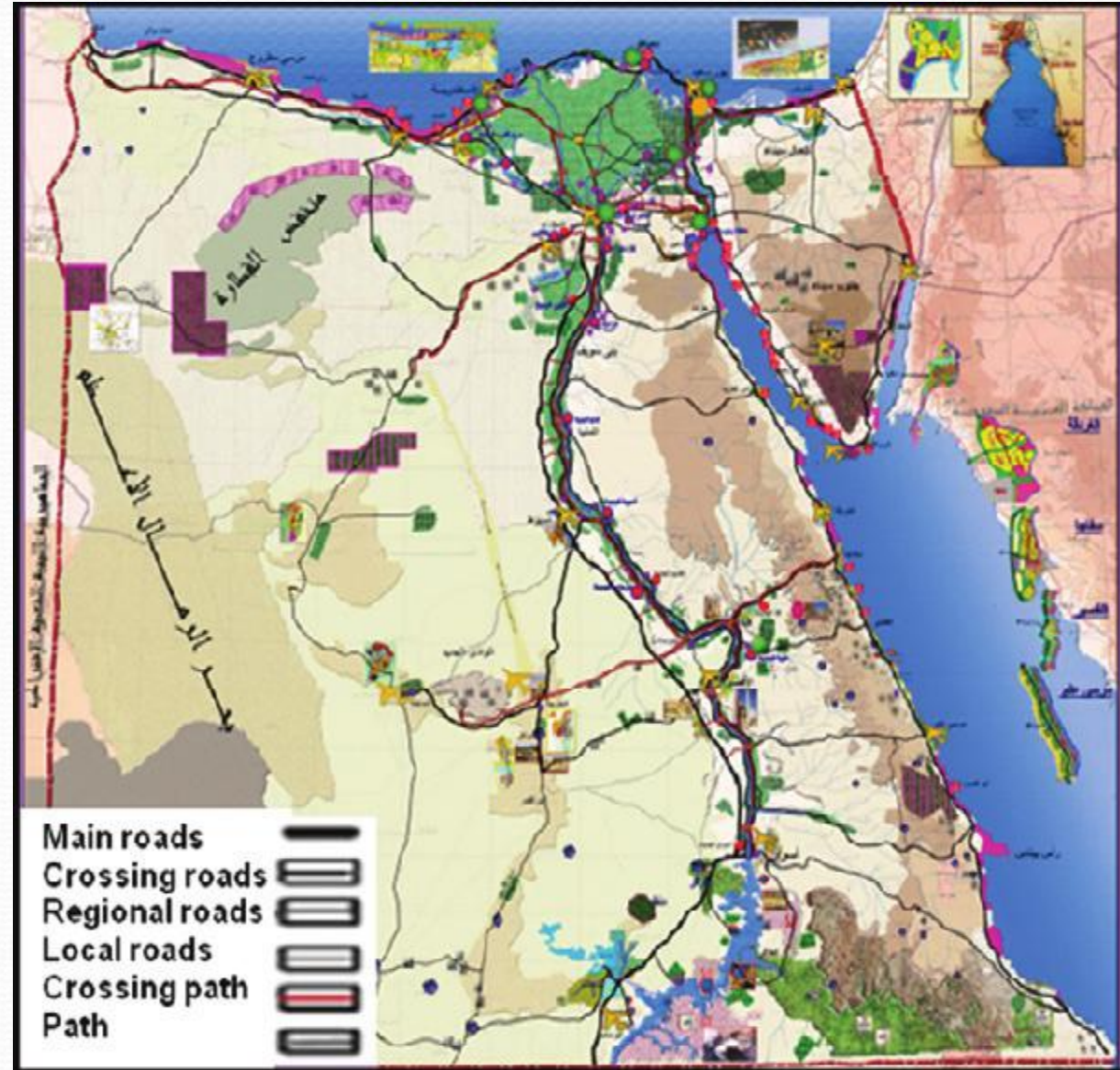
- a type of maps that provides detailed information about the *physical geography* and *natural features* of a region.
- They depict natural features such as mountains, plains, rivers, and oceans, water bodies, and vegetation.
- They utilize *colors* to differentiate between various *physical features* as well as the *elevations* of those *features*.



physical map of Egypt

2. Road Maps :

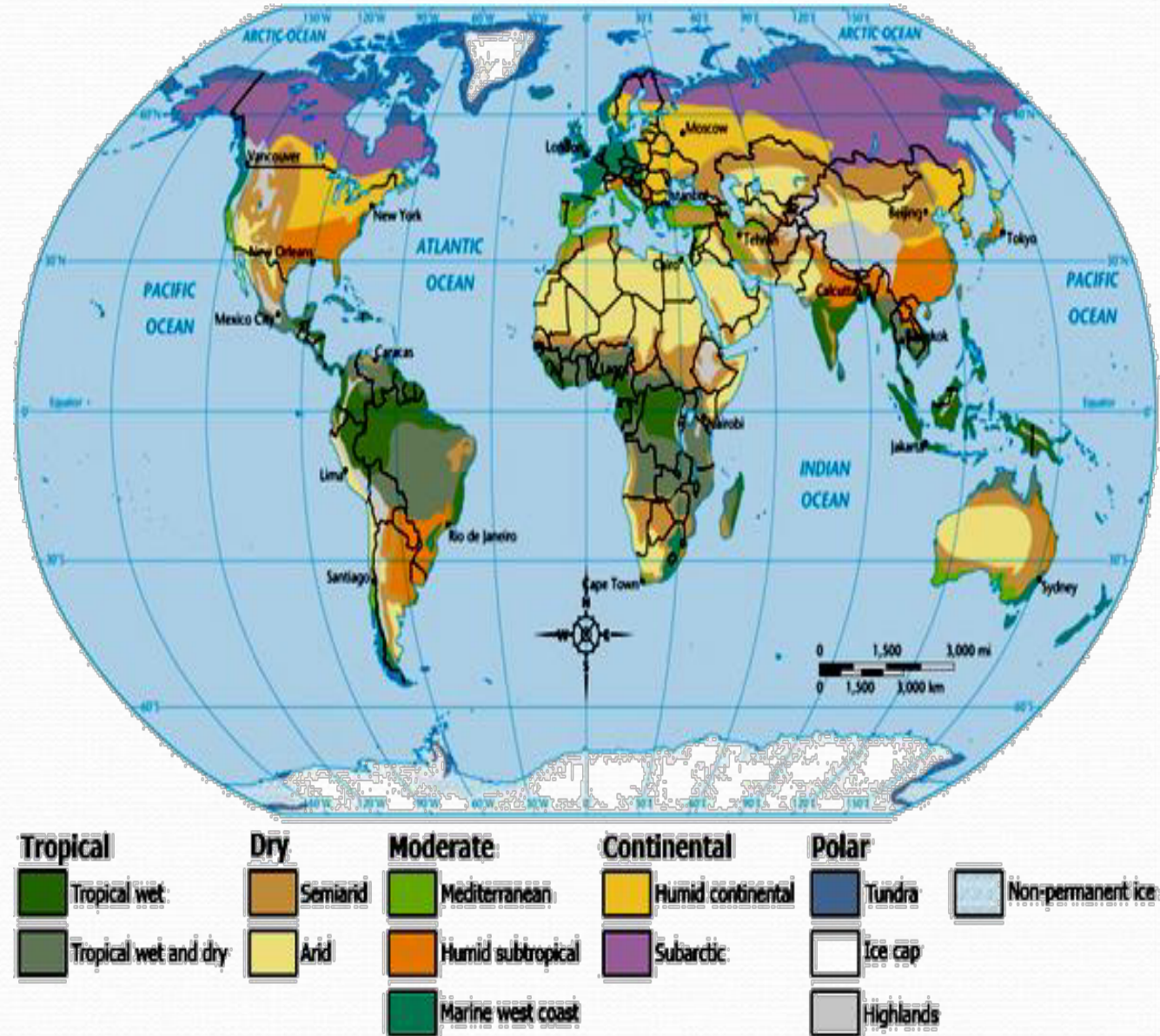
- *Road maps* are used to show *highways* and *roads* with varying levels of *detail*, as well as important *natural* and *manmade locations* such as *connecting cities* or *national parks*.
- Road maps also differentiate between the *sizes of roads*; for example, a *national highway* would likely be *wide* and *red*, whereas a *smaller road* would be *narrow* and *grey*.



Road map of Egypt

3. Climate Maps :

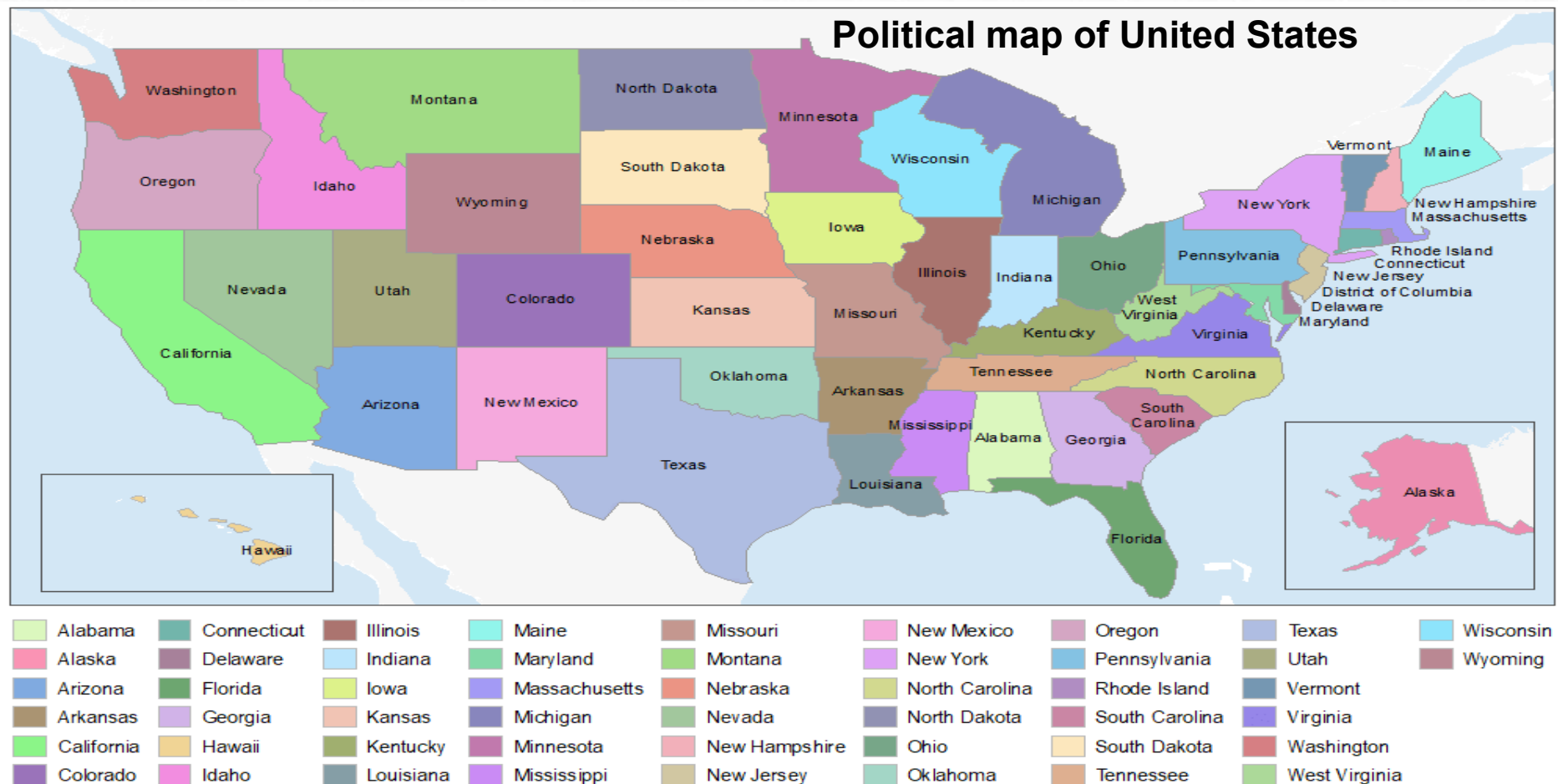
- *Climate maps* convey information about the *climate conditions* of an area, including **temperature** and **rainfall**.
- They are also usually *color-coded* to differentiate between various *climate zones*.



Climate map of the world

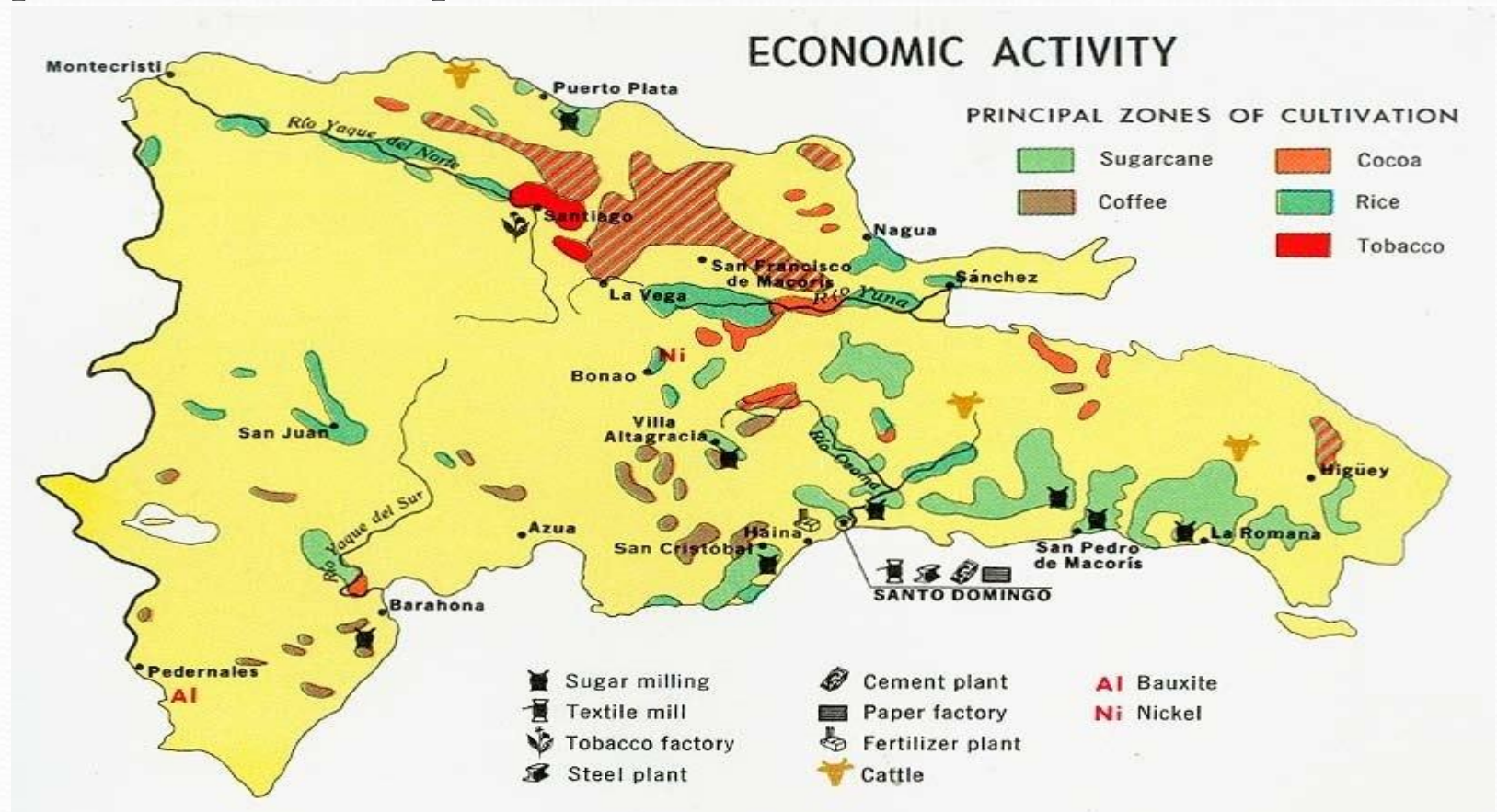
4. Political Maps :

- *Political maps* are used to display *artificial boundaries* such as *state* or *national borders*, as well as *cities* and sometimes *bodies of water*.
- *Political maps* are often *color-coded* to differentiate between different *countries* or *states*.



5. Economic Maps :

- *Economic maps* display the arrangement of *natural resources* and *economic activity* within a place.
- They are usually *color-coded* to differentiate between the various *resources* or *economic activity* present within the place.



Economic map of Dominican



End of Lecture_04